



Space Biology Evidence Engine

Citation-first research synthesis from a controlled corpus

John Hernandez | jherna65@fau.edu
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The problem

Space biology evidence is fragmented and difficult to synthesize safely

Scattered evidence

Relevant studies span publishers, organisms, experimental models, and terminology.

High trust bar

Scientific users need passage-level provenance, not unsupported model recall.

Slow comparison

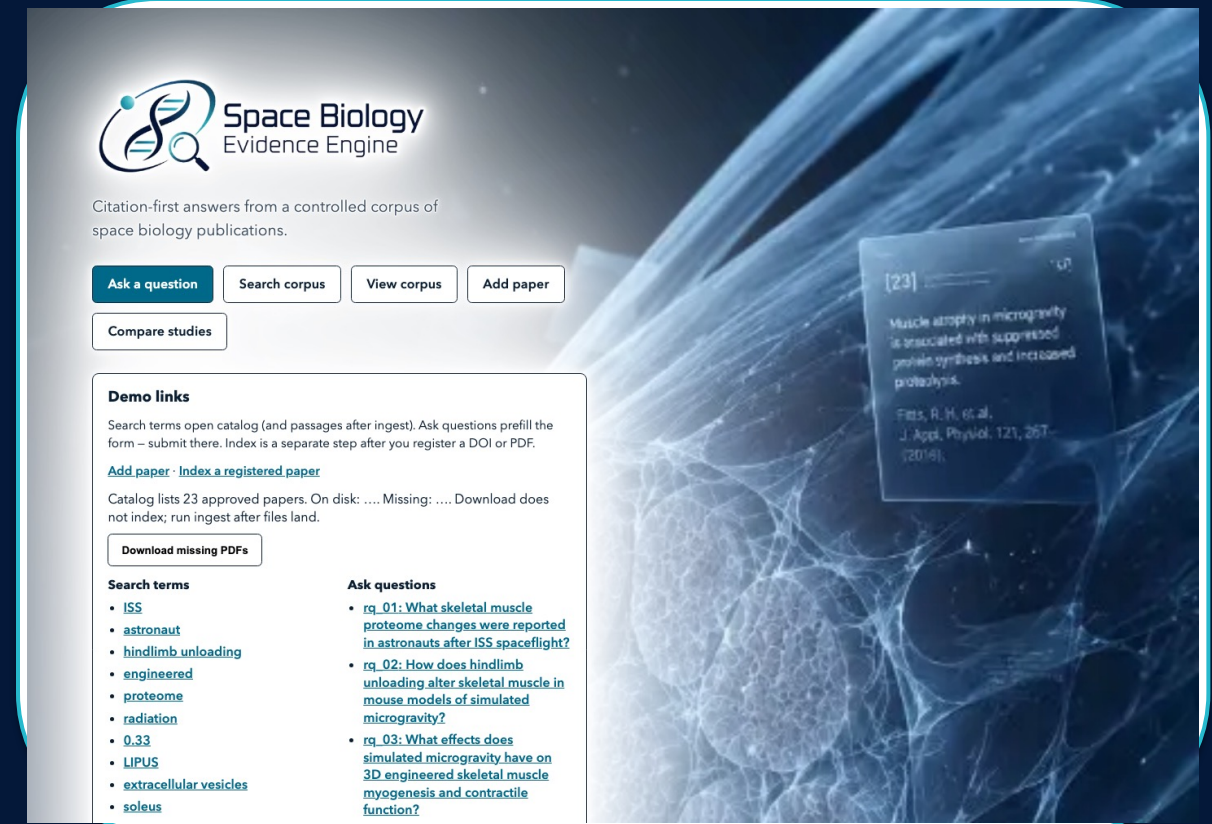
Researchers repeatedly search, read, and reconcile study context by hand.

- Primary users: space-biology researchers, students, educators, and corpus maintainers

The solution

A controlled, provenance-preserving evidence engine

- 23 owner-approved open-access publications
- Semantic + keyword retrieval over indexed passages
- Grounded answers with passage-level citations
- Explicit insufficient-evidence responses
- Local-first operation with optional OpenAI or Ollama generation



Architecture

Retrieval and generation remain independently testable



- Next.js frontend
- FastAPI API
- SQLAlchemy + Alembic
- Local embeddings
- Provider-neutral LLM interface

Grounded question answering

Answers are constrained to retrieved corpus passages



Ask a question

Submit a research question and receive a grounded answer drawn from the controlled corpus. Every claim should link to cited passages with publication, section, and page provenance.

Grounded question

The first block is the answer. Everything below it is supporting material: claims, warnings, and quoted PDF passages for each [C1] marker. Paper titles under a quote are provenance, not the evidence.

Research question

How does microgravity affect skeletal muscle proteome changes in astronauts?

Passages to retrieve 8

Ask question

Evidence sufficiency

Weak or empty retrieval returns a fixed insufficient-evidence response without invoking the LLM.

Citation validation

Unknown, missing, or invented citation markers fail closed instead of becoming successful answers.

Controlled corpus and provenance

Inventory visibility makes the evidence boundary explicit



Corpus

Metadata for the proposed August MVP corpus (microgravity and skeletal muscle). The 23 cards are **inventory**, not files on disk. Use **Download missing PDFs** on Corpus or Home if `data/pdfs/` is empty, then ingest. Compare inventory fields across studies. Add a local extra (DOI or PDF) – not one of the approved 23.

Catalog lists 23 approved papers. On disk: Missing: Download does not index; run ingest after files land.

[Download missing PDFs](#)

23 publications 17 CC BY 6 CC BY-NC-ND DOI links only

pub_001 2024 CC BY pending

Spaceflight on the ISS changed the skeletal muscle proteome of two astronauts.

human spaceflight not ingested

astronaut skeletal muscle proteome

[Publication details](#) [View at DOI](#)

pub_002 2023 CC BY pending

Genetic diversity modulates the physical and transcriptomic response of skeletal muscle to simulated microgravity in male mice.

mouse simulated microgravity not ingested

hindlimb unloading genetic diversity

[Publication details](#) [View at DOI](#)

pub_003 2024 CC BY pending

Simulated microgravity attenuates myogenesis and contractile function of 3D engineered skeletal muscle tissues.

engineered tissue simulated microgravity

not ingested

3D engineered skeletal muscle

[Publication details](#) [View at DOI](#)

pub_004 2023 CC BY pending

Myeloid cell infiltration in skeletal muscle after combined hindlimb unloading and radiation exposure in mice.

mouse hindlimb unloading radiation

not ingested

pub_005 2023 CC BY pending

Impacts of radiation exposure, hindlimb unloading, and recovery on murine skeletal muscle cell telomere length.

mouse hindlimb unloading radiation

not ingested

pub_006 2021 CC BY pending

Update on the effects of microgravity on the musculoskeletal system.

multi microgravity review not ingested

musculoskeletal review

[Publication details](#) [View at DOI](#)

[Back to corpus](#)

pub_001

Spaceflight on the ISS changed the skeletal muscle proteome of two astronauts.

2024 CC BY human spaceflight

Publication Metadata

Publication ID	pub_001
Title	Spaceflight on the ISS changed the skeletal muscle proteome of two astronauts.
DOI	10.1038/s41526-024-00406-3
Year	2024
License	CC BY
Organism	human
Exposure	spaceflight
Corpus note	astronaut skeletal muscle proteome
Approval status	pending
Ingestion status	not ingested

Source And Provenance

Original publication	Open publisher DOI page
Stored source URL	https://doi.org/10.1038/s41526-024-00406-3

Search and study comparison

Discovery without collapsing distinct evidence types

The screenshot shows the 'Search' page of the Space Biology Evidence Engine. At the top, the logo and title 'Space Biology Evidence Engine' are visible. Below the title, the word 'Search' is prominently displayed. A brief description states: 'Query stored publication metadata and any exposed passage records while keeping publication IDs, source links, sections, and pages visible.' The main search area includes a 'Search Stored Corpus' section with a 'Query' input field containing 'ISS' and a 'Search' button. Below this, it indicates '4 results for "ISS" (catalog only)'. The results are divided into two columns: 'Publications' and 'Passages'. The 'Publications' column shows two entries, each with a title, DOI, organism, exposure, and ingestion status. The first publication is 'Spaceflight on the ISS changed the skeletal muscle proteome of two astronauts.' and the second is 'Simulated microgravity attenuates myogenesis and contractile function of 3D engineered skeletal muscle tissues.' The 'Passages' column shows a message: 'No indexed passages yet. Catalog search still uses titles and labels. Run ingest, then keep the API running.' A 'View corpus' link is located at the top right of the search results area.

The screenshot shows the 'Compare studies' page of the Space Biology Evidence Engine. At the top, the logo and title 'Space Biology Evidence Engine' are visible. Below the title, the words 'Compare studies' are prominently displayed. A brief description states: 'Place corpus inventory records side by side. Organism and system categories stay labeled. This page does not synthesize unstated findings.' Below this, there are two tabs: 'Corpus' and 'Ask'. The 'Corpus' tab is selected. A text box explains: 'Comparison uses corpus inventory metadata only (organism, exposure, notes). It does not retrieve passages, generate findings, or invent differences that are not present in those fields. Organism classes stay labeled and are not merged.' Below this, there is a 'Select publications' section with the instruction 'Choose at least two records to compare.' A list of four publications is shown, each with a checkbox and a brief description. The first publication is 'pub_001' with the title 'Spaceflight on the ISS changed the skeletal muscle proteome of two astronauts.' and the description 'Human · human · spaceflight'. The second is 'pub_002' with the title 'Genetic diversity modulates the physical and transcriptomic response of skeletal muscle to simulated microgravity in male mice.' and the description 'Rodent · mouse · simulated microgravity'. The third is 'pub_003' with the title 'Simulated microgravity attenuates myogenesis and contractile function of 3D engineered skeletal muscle tissues.' and the description 'Engineered tissue · engineered tissue · simulated microgravity'. The fourth is 'pub_004' with the title 'Myeloid cell infiltration in skeletal muscle after combined hindlimb unloading and radiation exposure in mice.' and the description 'Rodent · mouse · hindlimb unloading radiation'. At the bottom, there is a button to 'Select two or more publications to open the comparison table.'

- Search preserves IDs, source links, sections, and pages
- Compare keeps organism and experimental-system labels visible

AI capabilities and safeguards

Useful automation with explicit scientific boundaries

Local embeddings

Sentence Transformers for private, low-cost semantic retrieval.

Provider-neutral generation

OpenAI and local Ollama paths behind a shared interface.

Hybrid retrieval

Vector similarity and PostgreSQL full-text retrieval with optional reranking.

Evidence gates

Minimum passage and publication support before generation.

Citation-first output

Passage IDs and publication provenance remain inspectable.

Hallucination checks

Offline fixtures flag unsupported claims and citation mismatches.

Results and engineering quality

A tested MVP with transparent limitations

23 publications

Approved controlled corpus inventory

292+ Python tests

Unit, API, retrieval, ingestion, citation, and accessibility coverage

CI quality gates

Ruff, Pyright, Python tests, Node typecheck and UI tests

- Citation correctness, hallucination, and retrieval evaluation harnesses
- No secrets committed; `.env` loading and fail-closed provider configuration
- Accessible interface patterns and keyboard-tested UI components

Demo readiness and operations

Local-first setup with a documented backup path

Primary demo

Run Postgres/pgvector, ingest approved PDFs, start FastAPI + Next.js, and use Ollama or OpenAI for grounded Ask.

Backup demo

Use repository screenshots and this deck if network, model, Docker, or showcase equipment is unavailable.

- Documented clean-machine setup and troubleshooting
- PDF fetch/ingest commands and provenance-preserving index
- Static corpus browsing remains available without the model provider

Known limitations and future work

The MVP is intentionally bounded

- Local deployment only; no public production URL
- Grounded Ask requires an indexed corpus plus configured OpenAI or Ollama provider
- Scientific outputs still require human review
- Corpus focuses on microgravity and skeletal muscle
- Future: stronger retrieval benchmarks, broader curated corpus, reviewer workflow, and production deployment
- Graph extraction remains experimental; no graph database in the MVP



Trustworthy evidence, one passage at a time

Space Biology
Evidence Engine

A practical foundation for citation-first space-biology research

- Controlled corpus
- Grounded retrieval
- Validated citations
- Transparent limitations

John Hernandez
jherna65@fau.edu
GitHub: [bluefate/spacebio-evidence-engine](https://github.com/bluefate/spacebio-evidence-engine)